



ELECTRICITY -07

Class 10 - Science

1. Using the following informations form a pathway to explain the flow of charges inside wire. And also include informations that are not mentioned below to complete it. Electron, solid conductor, solid crystal, drift speed, steady current [2]
2. What would be your explanation for your friend who thinks that electrons are consumed in an electric circuit? [2]
3. What is the direction of flow of current with respect to the flow of electrons in the circuit? [2]
4. State some properties of charge. [2]
5. You have three resistors of r ohm each and a battery of E volts. How would you connect these resistors with the battery to obtain maximum current? Draw a circuit diagram to illustrate your answer and also calculate the current drawn from the battery. [2]
6. What is an electric circuit? How are different components represented symbolically? [2]
7. How are conventional current and electronic current related ? [2]
8. What is meant by electric current? Write its SI unit. Calculate the amount of charge that flows through a conductor when a current 5A flows through it for 2 min. [2]
9. What will happen when : [2]
 - i. Voltmeter is connected in series?
 - ii. Ammeter is connected in parallel?
10. Write two points of difference between resistance and resistivity (or specific resistance). [2]
11. Two wires P and Q are made of copper. The wire P is long and thin, while the wire Q is short and thick. Which will have more specific resistance? Give a reason for your answer. [2]
12. A wire of given material having length l and area of cross-section A has a resistance of 4Ω . What would be the resistance of another wire of the same material having length $l/2$ and area of cross-section $2A$? [2]
13. What is electrical resistivity? In a series electrical circuit comprising of a resistor made up of a metallic wire, the ammeter reads 5 A. The reading of the ammeter decreases to half when the length of the wire is doubled. Why? [2]
14. A 4Ω resistance wire is doubled on it. Calculate the new resistance of the wire. [2]
15. Two metallic wires A and B are connected in series. Wire A has length l and radius r , while wire B has length $2l$ and radius $2r$. Find the ratio of total resistance of series combination and the resistance of wire A, if both the wires are of the same material? [2]
16.
 - a. List the factors on which the resistance of a conductor in the shape of a wire depends. [2]
 - b. Why are metals good conductors of electricity whereas glass is a bad conductor of electricity? Give reason.
 - c. Why are alloys commonly used in electrical heating devices? Give reason.
17. Why does resistance of a metallic conductor increase with increase in temperature? [2]
18. (a) Define electric resistance of a conductor? [2]

(b) A wire of length L and resistance R is stretched so that its length is double and the area of cross section is halved. How will its

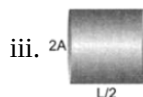
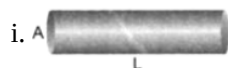
(i) resistance change?

(ii) resistivity change?

19. On what factors does the resistance of a conductor depend? [2]

20. A given length of a wire is doubled on itself. By what factor does the resistance of the wire change? [2]

21. Figure (a), (b) and (c) show three cylindrical copper conductors along with their face areas and length. Which of the conductors will have highest resistance and why? [2]



22. Why should a connection wire be thick? [2]

23. Will current flow more easily through a thick wire or a thin wire of the same material, when connected to the same source? Why? [2]

24. On what factors does the resistance of a conductor depend? [2]

25. Write the formula for current I flowing through a conductor if n electrons flow through the cross-section of a conductor in time t . [2]

26. i. Distinguish between the terms electrical resistance and resistivity of conductor. [2]

ii. A copper wire of resistivity $1.63 \times 10^{-8} \pi \text{ m}$ has cross-section area of $10.3 \times 10^{-4} \text{ cm}^2$. Calculate the length of the wire required to make a 20Ω coil.

27. The atoms of copper contain electrons and the atoms of rubber also contain electrons, then, why does copper conduct electricity but rubber does not conduct electricity? [2]

28. Which of the cables, one rated 5 A and the other 10 A will be of thicker wire? Give a reason for your answer. [2]

29. A copper wire has diameter 0.5 mm and resistivity $1.6 \times 10^{-8} \Omega \text{ m}$. What will be the length of this wire to make its resistance 10Ω ? How much does the resistance change if the diameter is doubled? [2]

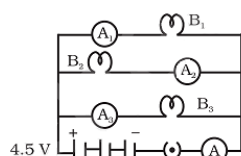
30. How does the use of fuse wire protect electrical appliances? [2]

31. Why does the connecting cord of an electric heater not glow while the heating element does? [2]

32. Though the same current flows through line wires and the filament of a bulb, yet only the latter glows. Why? [2]

33. Why does the cord of an electric heater not glow while the heating element does? [2]

34. B_1 , B_2 and B_3 are three identical bulbs connected as shown in figure. When all the three bulbs glow, a current of 3A is recorded by the ammeter A. [2]



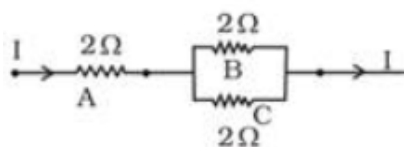
How much power is dissipated in the circuit when all the three bulbs glow together?

35. Alloys are used in electrical heating devices rather than pure metals. Give reason. [2]

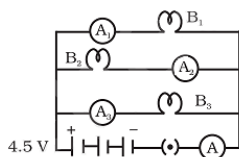
36. Should the heating element of an electric iron be made of iron, silver or nichrome wire? Justify giving three reasons? [2]

37. a. Write the mathematical expression for Joule's law of heating. [2]

- b. Compute the heat generated while transferring 96000 coulombs of charge in two hours through a potential difference of 40 V.
38. Why is the tungsten metal more coiled in the bulb and not installed in straight parallel wire form? [2]
39. State the energy conversion taking place in the following electric appliances: [2]
- Electric heater,
 - Electric motor,
 - Loud-speaker,
 - Electrolysis
40. a. Write Joule's law of heating. [2]
- b. Two lamps, one rated 100 W; 220 V, and the other 60 W; 220 V, are connected in parallel to electric mains supply. Find the current drawn by two bulbs from the line, if the supply voltage is 220 V.
41. Why is lead-tin alloy used for fuse wires? [2]
42. What is Joule's heating effect? How can it be demonstrated experimentally? List its four applications in daily life. [2]
43. Three $2\ \Omega$ resistors, A, B and C, are connected as shown in figure. Each of them dissipates energy and can withstand a maximum power of 18 W without melting. Find the maximum current that can flow through the three resistors? [2]



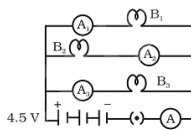
44. B_1 , B_2 and B_3 are three identical bulbs connected as shown in figure. When all the three bulbs glow, a current of 3A is recorded by the ammeter A. [2]



What happens to the reading of A_1 , A_2 , A_3 and A when the bulb B_2 gets fused?

45. 100 J of heat is produced each second in a $4\ \Omega$ resistance. Find the potential difference across the resistor. [2]
46. Why are the heating elements of electric toasters and electric irons made of an alloy rather than a pure metal? [2]
47. Mention two special features of the material used as an element of an electric iron. [2]
48. Two conducting wires of the same material and of equal lengths and equal diameters are first connected in series and then in parallel in an electric circuit. The ratio of heat produced in series and parallel combinations would be [2]
- 1:2
 - 2:1
 - 1:4
 - 4:1
49. What is meant by Joule's heating ? [2]
50. Why does an electric bulb become dim when an electric heater in parallel circuit is switched on? Why does dimness decrease after some time? [2]
51. Name three important effects produced by electric current. [2]
52. An electric heater of resistance $8\ \Omega$ draws 15 A from the service mains 2 hours. Calculate the rate at which heat is developed in the heater. [2]

53. Radhika is a student of class X. Her mother was making tea in an old electric kettle having metal case. When she switched on the power supply to the electric kettle. She got a severe electric shock. Radhika put off the main switch quickly and found that the connecting cord was torn, where her mother touched the metal case of the kettle. She also found that the red and black wires of connecting cord were firmly connected to the two lower terminals of the power plug but the green wire of cord was not connected to the upper terminal of the plug. Radhika replaced the torn connecting cord and also connected to the three wires of cord firmly to the power plug terminals. [2]
- On the basis of the above passage, answer the following questions:
- Why did Radhika put off the main switch quickly?
 - Which wire red, black or green, touched the metal case of electric kettle when Radhika's mother got electric shock?
 - What values are displayed by Radhika in this incident?
54. Compute the heat generated while transferring 96,000 coulomb of charge in one hour through a potential difference of 50V. [2]
55. Heat is generated continuously in an electric heater, but the temperature of its element becomes constant after some time. Why? [2]
56. A heater coil is rated 100 W, 200 V. It is cut into two identical parts. Both parts are connected together in parallel to the same source of 200 V. Calculate the energy liberated per second in the new combination. [2]
57. If the current I through a resistor is increased by 100% (assume that temperature remains unchanged), then find the increase in power dissipated. [2]
58. What is meant by electric power ? Give its units. How is it related to V and I ? [2]
59. Two lamps, one rated 100 W at 220 V, and the other 60 W at 220 V, are connected in parallel to electric mains supply. What current is drawn from the line if the supply voltage is 220 V? [2]
60. A lamp consumes 50 W and is lighted 2 h daily in month of April. How many units of electric energy is consumed ? [2]
61. An electric heater rated 1100 W operates at 220 V. Calculate (i) its resistance, and (ii) the current drawn by it. [2]
62. Which uses more energy, a 250 W TV set in 1 hr, or a 1200 W toaster in 10 minutes? [2]
63. An electric kettle is rated at (1000 W - 220 V). [2]
- What is the resistance of its element when in use?
 - What is the safe value of current that can pass through its element?
64. Calculate the cost of seeing 2 movies on colour T.V. daily for the month of September. [2]
Given wattage of colour T.V. = 60 W, duration each movie is 2 hours 30 min and 1kWh costs Rs. 4
65. You have two circuits: [2]
- a 6V battery is series with 1Ω and 2Ω resistors
 - a 4V battery in parallel with 12Ω and 2Ω . resistors
- Compare the power used in 2Ω resistor in each case.
66. Compare the power used in the 2Ω resistor in each of the following circuits: [2]
- a 6 V battery in series with 1Ω and 2Ω resistors, and
 - a 4 V battery in parallel with 12Ω and 2Ω resistors.
67. B_1 , B_2 and B_3 are three identical bulbs connected as shown in figure. When all the three bulbs glow, a current of 3A is recorded by the ammeter A. [2]



What happens to the glow of the other two bulbs when the bulb B_1 gets fused?

68. What determines the rate at which energy is delivered by a current? [2]
69. A potential difference of 220 V is applied across a resistance of $440\ \Omega$ in an electric ion. [2]
(i) Find the current.
(ii) Heat energy produced is 30s.
70. Write two points of difference between electric energy and electric power. [2]
71. Several electric bulbs designed to be used on a 220V electric supply line are rated 10W. How many lamps can be [2]
connected in parallel with each other across the two wires of 220V line if the maximum allowable current is 5A?
72. An electric motor takes 5A from a 220V line. Determine the power of the motor and the energy consumed in 2 [2]
h.
73. An electric iron is rated **1 kW - 220 V**. Calculate the following: [2]
i. The resistance of its heating element.
ii. The amount of current that will flow through the element.
iii. The amount of heat that will be produced in 2 minutes.
iv. The power consumed if the line voltage falls to 200 V.
74. An electric kettle rated at 220 V, 2.2 kW works for 3h. Calculate the energy consumed and the current drawn. [2]
75. Suppose your parents have constructed a two room house and you want that in the living room there should be a [2]
provision of one electric bulb, one electric fan, a refrigerator and a plug point for appliances of power up to 2
kilowatt. Draw a circuit diagram showing electric fuse and earthing as safety devices.